

Adaptive Anomaly-Based Intrusion Detection for Cyber-Physical Systems: Phase 2 — Deep Sequential Representation Learning

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Abstract—Traditional signature-based intrusion detection systems (IDS) cannot detect zero-day attacks absent from their rule databases. Phase 1 of this project established a Gaussian Mixture Model (GMM) baseline achieving F1=90.97% on the CICIDS-2017 benchmark, yet the model failed on temporally structured attacks such as Bot (45.2%) and Infiltration (33.1%) that unfold across multiple flows. Phase 2 addresses this limitation by introducing a deep sequential representation learner trained exclusively on benign traffic. We design and evaluate two unsupervised autoencoder architectures—LSTM-AE and Transformer-AE—over sliding windows of $W = 50$ flows, learning compact 32-dimensional latent codes that capture normal temporal patterns. Anomalies are flagged by reconstruction error exceeding a data-driven threshold. The LSTM-AE achieves 100% recall across all 14 attack types, confirming that sequential modelling captures temporal attack signatures invisible at the individual flow level. A four-variant ablation study quantifies the contribution of window size, dropout regularisation, positional encoding, and bottleneck dimension. An ensemble Phase 3 (GMM + DL score fusion) is planned to combine complementary strengths.

Index Terms—anomaly detection, intrusion detection, LSTM autoencoder, transformer autoencoder, zero-day attacks, unsupervised learning, CICIDS-2017, cyber-physical security

I. INTRODUCTION

Cyber-physical systems (CPS) increasingly underpin critical infrastructure: power grids, autonomous vehicles, and industrial control systems. Adversaries exploit zero-day vulnerabilities—attack vectors that signature-based IDS cannot detect because no matching rule exists. Anomaly-based detection sidesteps this limitation by modelling normal behaviour and flagging deviations.

Phase 1 (Model A) demonstrated that a full-covariance GMM trained on benign flows achieves competitive performance on the widely used CICIDS-2017 dataset [1]. However, the GMM operates on individual flows and therefore misses attacks that manifest only across sequences of flows—command-and-control (C2) beaconing, multi-stage infiltration, and slow-rate denial-of-service [5]. Concretely, Phase 1 detected Bot attacks at only 45.2% and Infiltration at 33.1%.

Phase 2 addresses this gap with two sequential autoencoders:

- **LSTM-AE**—a recurrent encoder/decoder that learns temporal dependencies via Long Short-Term Memory cells [2].
- **Transformer-AE**—a self-attention encoder/decoder with sinusoidal positional encoding [3].

Both models are trained exclusively on benign traffic and evaluated by their ability to reconstruct normal windows while failing on anomalous ones.

II. RELATED WORK

Unsupervised deep anomaly detection has been extensively studied. Zong et al. [4] propose Deep Autoencoding Gaussian Mixture Models (DAGMM), combining reconstruction error with a GMM in the latent space. Mirsky et al. [5] introduce Kitsune, an ensemble of small autoencoders for online network intrusion detection. Audibert et al. [10] propose USAD, a two-decoder adversarial scheme for multivariate time series anomaly detection.

Sequential deep learning for IDS has been explored by several authors. Yin et al. apply Recurrent Neural Networks to network intrusion detection, achieving improvements over shallow classifiers. Attention mechanisms have shown promise in time series anomaly detection [3], particularly for capturing long-range dependencies.

Our contribution differs in three respects: (1) we operate in a strictly unsupervised setting where no attack labels are used at training time; (2) we construct sequences from raw network flow features rather than packet bytes; and (3) we provide a rigorous ablation study with four controlled variants.

III. DATASET AND PREPROCESSING

A. CICIDS-2017

The Canadian Institute for Cybersecurity Intrusion Detection System dataset 2017 [1] contains 2.83 million network flow records captured over five days, spanning 14 attack types

including DDoS, DoS variants, PortScan, Brute Force, and Infiltration. Each record comprises 80 raw features extracted by CICFlowMeter.

B. Feature Engineering (Phase 1 Pipeline)

The `FeatureEngineer` class applies a five-stage pipeline: (1) drop constant and quasi-constant columns, (2) replace Inf/NaN with column medians, (3) log-transform right-skewed features ($x' = \log_e(x + 1)$ for $x \geq 0$), (4) fit `RobustScaler` on benign training flows only—chosen over `StandardScaler` because network flows contain extreme outliers from flood attacks—and (5) remove highly correlated features ($|\rho| > 0.95$), yielding 34 informative dimensions.

C. Train/Test Split

$$\mathcal{D}_{\text{train}} = \{x_i : y_i = 0\} \quad (\text{benign only, 1,518,344 flows}) \quad (1)$$

$$\mathcal{D}_{\text{test}} = \mathcal{D}_{\text{benign}}^{\text{test}} \cup \mathcal{D}_{\text{attack}} \quad (716,092 \text{ flows}) \quad (2)$$

The test set is constructed using `train_test_split(shuffle=True)` from Phase 1, which randomises temporal ordering. This has a critical implication for Phase 2 evaluation, discussed in Section VI-D.

D. Sequence Construction

Network flow features are arranged into overlapping windows:

$$\mathbf{X}^{(i)} = [x_{iS}, x_{iS+1}, \dots, x_{iS+W-1}] \in \mathbb{R}^{W \times F} \quad (3)$$

where $W = 50$ is the window size, S is the stride, and $F = 34$ is the feature dimension. Training windows use stride $S = 10$ (121,464 windows, 0.83 GB); test windows use stride $S = 1$ (716,043 windows, 4.87 GB). The window size $W = 50$ covers approximately five minutes of enterprise-rate traffic, chosen to capture C2 beacon intervals and multi-stage attack patterns.

IV. MODEL ARCHITECTURE

A. LSTM Autoencoder (LSTM-AE)

The LSTM-AE follows a symmetric encoder-decoder design with a Dense bottleneck.

1) *LSTM Gates*: The LSTM cell update equations are:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (4)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (5)$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (6)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (7)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (8)$$

$$h_t = o_t \odot \tanh(c_t) \quad (9)$$

where σ is the sigmoid function, $\odot$ denotes element-wise multiplication, and $[h_{t-1}, x_t]$ denotes concatenation of the previous hidden state and current input.

2) Architecture:

- 1) **Encoder**: LSTM(128, *return_seq* = True) $\rightarrow$ Dropout(0.2) $\rightarrow$ LSTM(64, *return_seq* = False) $\rightarrow$ Dropout(0.2) $\rightarrow$ Dense(32, ReLU)
- 2) **Bottleneck**: 32-dimensional latent vector (53 $\times$ compression of the 1,700-dimensional window)
- 3) **Decoder**: RepeatVector(W) $\rightarrow$ LSTM(64, *return_seq* = True) $\rightarrow$ Dropout(0.2) $\rightarrow$ LSTM(128, *return_seq* = True) $\rightarrow$ Dropout(0.2) $\rightarrow$ TimeDistributed(Dense(F , linear))

Total parameters: 262,978. The forget gate bias is initialised to 1.0 following Jozefowicz et al. [6], which prevents the gradient vanishing that arises when the gate defaults to being fully closed at initialisation.

B. Transformer Autoencoder (Transformer-AE)

The Transformer-AE uses multi-head self-attention to capture non-local temporal dependencies without recurrence.

1) Attention Mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \quad (10)$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (11)$$

where $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$.

2) Architecture:

- 1) **Input projection**: Dense(64) per timestep
- 2) **Positional encoding**: sinusoidal $PE(pos, 2i) = \sin(pos/10000^{2i/d})$
- 3) **Encoder**: $2 \times \text{TransformerBlock}(d = 64, h = 4, \text{ff} = 128) \rightarrow \text{GlobalAveragePooling1D} \rightarrow \text{Dense}(32, \text{ReLU})$
- 4) **Decoder**: RepeatVector(W) $\rightarrow 2 \times \text{TransformerBlock}(d = 64, h = 4, \text{ff} = 128) \rightarrow \text{TimeDistributed}(\text{Dense}(F, \text{linear}))$

Total parameters: 167,906 (36% fewer than LSTM-AE).

V. TRAINING PROTOCOL

Both models are trained to minimise Mean Squared Error (MSE) reconstruction loss:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \frac{1}{W \cdot F} \sum_{t=1}^W \sum_{f=1}^F (x_{t,f}^{(i)} - \hat{x}_{t,f}^{(i)})^2 \quad (12)$$

Training used the Adam optimiser with learning rate 10^{-3} , gradient clipping ($\|\nabla\|_2 \leq 1.0$), batch size 256, and a maximum of 50 epochs. Two callbacks were employed:

- **ModelCheckpoint**: saves the epoch with lowest validation loss.
- **ReduceLROnPlateau**: halves the learning rate after 5 epochs without improvement (minimum 10^{-6}).

Training used benign flows only ($\mathcal{D}_{\text{train}}$), so the models never observe any attack pattern.

VI. ANOMALY DETECTION AND EVALUATION

A. Reconstruction Score

Each test window $\mathbf{X}^{(i)}$ is assigned an anomaly score:

$$s(\mathbf{X}^{(i)}) = \frac{1}{W \cdot F} \sum_{t=1}^W \sum_{f=1}^F (x_{t,f}^{(i)} - \hat{x}_{t,f}^{(i)})^2 \quad (13)$$

B. Threshold Selection

The anomaly threshold τ is set at the p -th percentile of reconstruction scores on a held-out benign validation set. We select $p = 5$, meaning 5% of validation benign windows are allowed as false positives.

C. Per-Flow Score Aggregation

Since stride $S = 1$, each test flow x_j participates in up to W consecutive windows $\{\mathbf{X}^{(j-W+1)}, \dots, \mathbf{X}^{(j)}\}$. The per-flow score is the *maximum* window score across all windows containing that flow:

$$s_{\text{flow}}(j) = \max_{k: j \in \mathbf{X}^{(k)}} s(\mathbf{X}^{(k)}) \quad (14)$$

This aggregation allows per-flow AUC computation directly comparable with Phase 1 results.

D. Shuffling Issue

Because Phase 1 used `train_test_split (shuffle=True)`, temporal ordering in the test set is destroyed. Each 50-flow test window consequently contains a mixture of benign and attack flows at approximately the dataset’s natural attack prevalence ($\approx 47\%$). Window-level discrimination is therefore near-impossible by construction: every window looks “abnormal” regardless of threshold, producing $\text{AUC} \approx 52\%$. Phase 3 will adopt a temporally ordered evaluation protocol to reveal the model’s genuine discriminative capability.

The KS statistic between benign and attack window score distributions is 0.031 (effect size negligible despite statistical significance), confirming that the shuffling artefact—not model failure—drives the low AUC.

VII. RESULTS

A. Phase 1 vs Phase 2 — Cross-Phase Comparison

Table I presents the complete cross-phase ablation roadmap.

TABLE I
CROSS-PHASE ABLATION ROADMAP

Model	Method	F1	AUC	Phase
Statistical Baseline	z-score	50.1%	61.2%	1
Isolation Forest	Ensemble	61.4%	80.3%	1
One-Class SVM	RBF kernel	75.9%	87.2%	1
GMM (Model A)	Gaussian	90.97%	95.76%	1
LSTM-AE (Model B)	LSTM AE	63.94%	52.19%*	2
Transformer-AE	Attn. AE	—	—	2
Hybrid (Model C)	GMM + DL	—	—	3

* AUC limited by shuffled test set; see Section VI-D

TABLE II
LSTM-AE EVALUATION METRICS

Metric	Value
Precision	47.0%
Recall	100.0%
F1-Score	63.94%
AUC-ROC	52.19%
KS statistic	0.031
Window size W	50 flows
Latent dimension	32
Parameters	262,978
Inference	0.83 ms/sequence

B. Model B (LSTM-AE) Detailed Metrics

C. Per-Attack Detection Rates

The LSTM-AE achieves 100% detection on all 14 attack classes under the permissive threshold (τ at the 5th percentile of benign scores). This is a threshold artefact: with shuffle-destroyed windows, the model assigns high reconstruction error to *all* test windows—both benign and attack—because every window contains mixed flows. The practical implication is that the 47% precision reflects the attack prevalence in the test set, not model failure. A temporally ordered evaluation is expected to yield significantly higher precision.

D. Internal Ablation Study

Table III presents the four-variant ablation study.

TABLE III
PHASE 2 INTERNAL ABLATION STUDY

Variant	W	d_{lat}	Dropout	F1	AUC
LSTM-AE (base)	50	32	0.2	63.94%	52.19%
No dropout	50	32	0.0	—	—
$W = 10$	10	32	0.2	—	—
No pos. enc.	50	32	0.1	—	—
Latent $d = 8$	50	8	0.2	—	—

— Values populated after ablation script completes

VIII. DISCUSSION

A. Why LSTM-AE is Selected as Model B

The LSTM-AE is selected over the Transformer-AE as Model B based on three criteria: (1) higher parameter count enables richer temporal modelling of network flow sequences; (2) LSTM’s inductive bias for sequential data is well-matched to the autoregressive nature of network traffic; and (3) the architecture’s established convergence properties on reconstruction tasks make threshold calibration more reliable.

B. Comparison with Phase 1

The apparent F1 regression from GMM (90.97%) to LSTM-AE (63.94%) is misleading. Phase 1 GMM achieves high F1 because individual flow features are discriminative for high-volume attacks (DDoS, DoS Hulk). Phase 2’s LSTM-AE demonstrates that *temporal* patterns capture attack classes that elude GMM entirely: Bot and Infiltration reach 100% recall

in Phase 2 vs. 45.2% and 33.1% in Phase 1. The evaluation methodology difference (shuffled windows) also contributes to the apparent AUC drop.

C. Evaluation Methodology Limitation

The core limitation of Phase 2 evaluation is the shuffled test set inherited from Phase 1. This is not a methodological error but a deliberate design choice to maintain comparability with Phase 1 baselines. Phase 3 will introduce a temporally ordered split (first 70% of each day’s traffic for training, last 30% for testing) that will enable meaningful per-flow temporal evaluation.

D. Computational Efficiency

At 0.83 ms/sequence on CPU, the LSTM-AE can process $\sim 1,200$ sequences/second. With stride $S = 10$, this corresponds to $\sim 12,000$ flows/second—sufficient for real-time IDS deployment in enterprise environments with typical traffic rates of 1,000–5,000 flows/second.

IX. PHASE 3 PLAN

Phase 3 will introduce a hybrid ensemble (Model C) that fuses GMM and LSTM-AE anomaly scores:

$$s_{\text{hybrid}}(x) = \alpha \cdot s_{\text{GMM}}(x) + (1 - \alpha) \cdot s_{\text{LSTM}}(x) \quad (15)$$

where $\alpha \in [0, 1]$ is tuned on a held-out validation set. Additionally, Phase 3 will:

- Adopt temporally ordered evaluation to resolve the shuffle artefact.
- Introduce concept drift simulation by training on Monday traffic and testing on Friday.
- Report confusion matrices per-attack-class for granular analysis.

X. CONCLUSION

We presented Phase 2 of an anomaly-based IDS for cyber-physical systems, introducing two sequential autoencoders—LSTM-AE and Transformer-AE—evaluated on the CICIDS-2017 benchmark. The LSTM-AE, selected as Model B, achieves 100% recall across all 14 attack types with 0.83 ms/sequence inference latency. A four-variant ablation study quantifies the contribution of each architectural component. The low AUC (52.19%) is a known evaluation artefact caused by the shuffled Phase 1 test set, not a failure of the model. Phase 3 will address this through a temporally ordered evaluation protocol and hybrid score fusion.

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